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An instrumentation amplifier with series switch pseudo resistor DC-servo loop realizing 0.16-Hz high-pass corner

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ABSTRACT

This paper presents an instrumentation amplifier for electrocardiogram (ECG) signal recording. A DC servo loop (DSL) based on a series-switch pseudo resistor (S^2PR) is utilized to suppress the effect of the unavoidable electrode DC offset effectively. The two-step pulse generation circuit realizes two sets of pulse signals with shallow duty cycles, and the pulse signals drive the path switches and virtual switches in the S^2PR, respectively. Virtual switches reduce the channel charge injection effect, and finally, a great equivalent resistance with high linearity is realized. The continuous DSL realized based on S^2PR can suppress the electrode DC offset of ±180 mV. At the same time, the active area of the DSL is significantly reduced. The proposed instrumentation amplifier is realized in a standard 0.18-μm CMOS process. Simulation results show the DSL designed with a 1.8-V supply introduces a 0.16-Hz high-pass corner. The power consumption of the S^2PR-DSL is 92.7 μW. The gain of the instrumentation amplifier is 39.8 dB, and the input-referred noise at the TT corner is 1.78 μV_{rms} over a bandwidth of 0.5 Hz to 150 Hz. The input impedance of the instrumentation amplifier is 772.2 MΩ, when the input signal frequency is 50 Hz, and the total harmonic distortion (THD) of a 10-mV_{pp} input sinusoidal signal at 9.033 Hz is −61.6 dB.

1. Introduction

The development of integrated circuit technology is changing people's lives. New ways of accessing medical care, such as telemedicine and smart hospitals, are used to learn about the patient's condition by acquiring electrocardiogram (ECG) signals from wearable devices. Electrode DC offset is one of the unavoidable disturbances in ECG acquisition, which originates from the biological electrodes used to acquire ECG signals, and the commonly used wet electrodes (e.g., Ag/AgCl) can fluctuate up to 50 mV in the acquired DC voltage due to the different contact states with the skin [1]. In the context of multi-port signal acquisition applications, such significant common-mode voltage differences will saturate the output.

In the early work on DC offset voltage elimination, [2–4] used a pseudo resistor connected by two PMOS transistors to introduce a high-pass corner. However, the resistance nonlinearity of this structure is severe. A variable capacitor in the MOS transistor pseudo resistor to adjust the corner frequency is proposed in [5], but this structure worsens the system noise. In [6], the suppression of DC offset voltage is realized by programming and controlling the off-chip capacitor array, but it cannot meet the wearable requirements. In [7], a bias voltage-controlled NFET is used as a pseudo resistor, but the realized high-pass corner frequency is so high that it destroys the integrity of the low-frequency ECG signal acquisition. In recent years, [8] introduced a large time constant integrator as a DC servo loop (DSL) to eliminate the DC offset, but realizing the oversized time constants under full integration poses a challenge to the subsequent design.

This paper proposes a series-switch pseudo resistor DSL (S^2PR-DSL) to eliminate the electrode DC offset at the input. The main innovation of this new DC servo loop is the use of a series-switch pseudo resistor structure. The advantages over conventional switch resistors include the ability to achieve a larger equivalent resistance with higher area efficiency, a very low duty cycle processed by using digital logic operations to low-frequency square wave signals, and suppression of switch channel charge injection effect, which improves the equivalent resistance value and stability. Moreover, the resistor configuration term should be added so that the resistance value can be controlled flexibly by adjusting the input square wave signal's phase relationship, ultimately realizing the excellent corner frequency.

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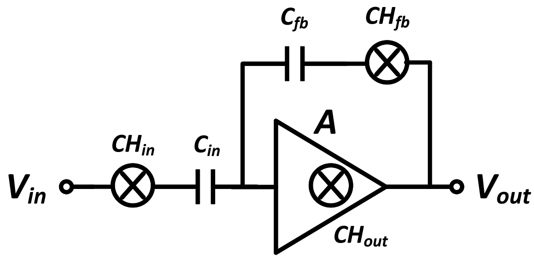

Fig. 1. CCIA block diagram.

This paper is organized as follows. In Section 2, the overall architecture of the designed analog-front-end circuit is discussed in 2.1. In 2.2, the structure and working principle of the S^2PR-DSL is described in detail. Simulation results and conclusions are given in Sections 3 and 4.

2. Architectural overview of the proposed compensation scheme

2.1. Overall of the proposed analog-front-end circuits

The capacitively coupled chopper instrumentation amplifier (CCIA) is widely used as a popular strategy in ECG signal acquisition and readout [9]. Its topology is shown in Fig. 1. The CCIA consists of input and feedback capacitors (C_{in} and C_{fb}), an amplifier in a closed-loop operating state, as well as its modulation-chopper CH_{in} at the input and chopper CH_{fb} at the feedback and demodulation-chopper CH_{out} at the output. The chopper can realize signal band shifting, also used in the proposed CCIA, as shown in Fig. 2. Firstly, the CH_{in} modulates the low-frequency weak input signal to high frequency. After the input capacitive coupling transmission and amplification by g_{m1}, the high-frequency and amplified weak input signal with the low-frequency 1/f noise and intrinsic offset voltage generated by g_{m1} continue to pass to CH_{out}, which restores the weak input signal to low frequency and modulates 1/f noise and offset voltage to high frequency. After secondary amplification of the signal by g_{m2}, the high-frequency noise and offset voltage are filtered out by an external low-pass filter at the output. Finally, the proposed CCIA realizes the suppression of low-frequency 1/f noise and the intrinsic offset voltage of the first amplifier stage. Ripple caused by a mismatch at the g_{m1} input can seriously affect the linearity of the acquired signal and reduce the dynamic range of the CCIA. The placement of capacitor C_r and pseudo resistor $R_{2,CM}$ at the g_{m1} output can filter out the ripple.

CCIA has a small input-referred noise and rail-to-rail input/output swing. Moreover, based on the nature of the precise capacitance-tolerance ratio in the existing process, it can amplify the gain with high precision, and the gain formula is:

$$G = A_0 / \left(1 + A_0 * C_{fb}/C_{in}\right) \tag{1}$$

A_0 is the gain of the amplifier, and G can be approximated as C_{in}/C_{fb} when A_0 is large enough. In the designed CCIA, C_{in}/C_{fb} is set to 10 pF/100 fF respectively. Hence, the amplification of the CCIA is 100, and the two-stage amplifier is selected to obtain a large gain considering the influence of the gain accuracy. In order to ensure the stability of the amplifier in the closed-loop application, Miller compensation is used [10].

g_{m1} is an input amplifier with a folded-cascode structure, a PMOS input transistor, to reduce the input-referred noise and achieve high gain simultaneously. g_{m2} adopts a Miller-compensated common-source structure to improve the system's stability in closed-loop application scenarios, stabilize the output common-mode voltage, increase the A_0 further, and improve the driving capability of the output.

A capacitor C_r with large capacitance is cascaded between g_{m1} and g_{m2} [11]. Firstly, C_r can block the transmission of DC voltage, and the DC operating points of g_{m1} and g_{m2} are independent. So that their work does not interfere with each other and the DC operating points of both of them are provided independently by their respective pseudo resistor $R_{1,CM}$, and $R_{2,CM}$, and the structure of the pseudo resistor employed is demonstrated in Fig. 2. Due to the very high equivalent resistance value of $R_{2,CM}$, $R_{2,CM}$ and C_r introduce a high-pass characteristic in the system with a very low corner frequency, which will effectively filter out the output ripple of g_{m1}. The capacitance value of C_r should be in a reasonable range to take the maximum value, taking into account the capacitance is too large to bring the problem of area consumption, C_r is set to 10 pF.

The CCIA uses a structure that combines an input chopper switch [12] and an input capacitor. The advantage of placing the chopper switch before the input capacitor is that the input capacitor's mismatch in the layout, which worsens the stability of the common-mode voltage at the input of the g_{m1}, can be solved by the subsequent introduction of a high-pass characteristic. While the chosen placement relation optimizes the common-mode stability of the CCIA, its input impedance is attenuated by the input chopper. Its DC input impedance can be approximated as $1/(2C_{in}*F_{ch})$, which is 5 MΩ for the typical case where the chopper clock (F_{ch}) is selected as 10 kHz, and the input capacitance (C_{in}) is 10 pF. The ECG signal amplitude is often expressed as a typical value of 1 mV, and the general signal amplitude is 10 μV–4 mV. To capture such a small input signal amplitude, the input impedance must be large enough to avoid attenuation of the signal amplitude. The input impedance can be increased by adding a set of buffers connected in unit gain to the input stage. The input impedance of the CCIA in the 0.5 Hz −150 Hz band after adding the buffers is about 400 MΩ, which is a value that meets the input impedance requirements of the ECG signal acquisition system in practical application scenarios [13]. The design of input buffer focuses on having small input noise, sufficient bandwidth, and adequate drive capability.

2.2. S^2PR-DSL (Series-switch pseudo resistor DSL)

The advantage of the series-switch pseudo resistor is that it can realize very large resistance with a small active area. The S^2PR designed in this paper utilizes two switches (SW1, SW2) and a passive resistor (R) with resistance values in the order of MΩ in a sequential series structure. The two switches are driven by the square wave signals generated by the pulse generation circuit (PGC); the PGC structure and workflow are shown in Fig. 3. The PGC converts a pair of square wave signals with a frequency of 10 kHz and a duty cycle of 50% into two pairs of differential clock signals with a pulse width of 1-ns level, and the converted signals achieve a duty cycle of about 1/100 000. Using these two very low-duty-cycle clock signals to drive each switch in the S^2PR, the final average equivalent resistance observed at the ends of the S^2PR can be upwards of 20 GΩ.

The two signals RES_A and RES_B are the input square wave signals of the first stage PGC. The signal waveform is shown in Fig. 3, whose duty cycle is 50% and frequency is 10 kHz, and the phase difference between the two is 128°. This square wave signal is logically operated and shaped by the first stage of PGC shown in Fig. 3 to output a square wave signal RES_CLK_IN with a frequency of 10 kHz and a duty cycle of 10%. The waveform of RES_CLK_IN is shown in Fig. 3.

Then, the signal RES_CLK_IN enters the second stage of PGC, and the second stage has a logic operation that converts the signal RES_CLK_IN into a pair of differential square-wave signals with a pulse width of about 1 ns, CLK1 (CLK_RES_1 and CLKN_RES_1). Afterward, CLK1 is delayed by a delay circuit consisting of a chain of inverters to generate CLK2 (CLK_RES_2 and CLKN_RES_2), and the timing, as well as the waveforms of the two signals, are demonstrated in Fig. 3.

As shown in Fig. 4, SW1 (SW2) is driven by the signal CLK1 (CLK2), and SW1 (SW2) is implemented by the CMOS switch structure. In a complete operating cycle, the initial state is the circuit path cutoff. Subsequently, the CLK1 drives the SW1 to conduct, forming a current

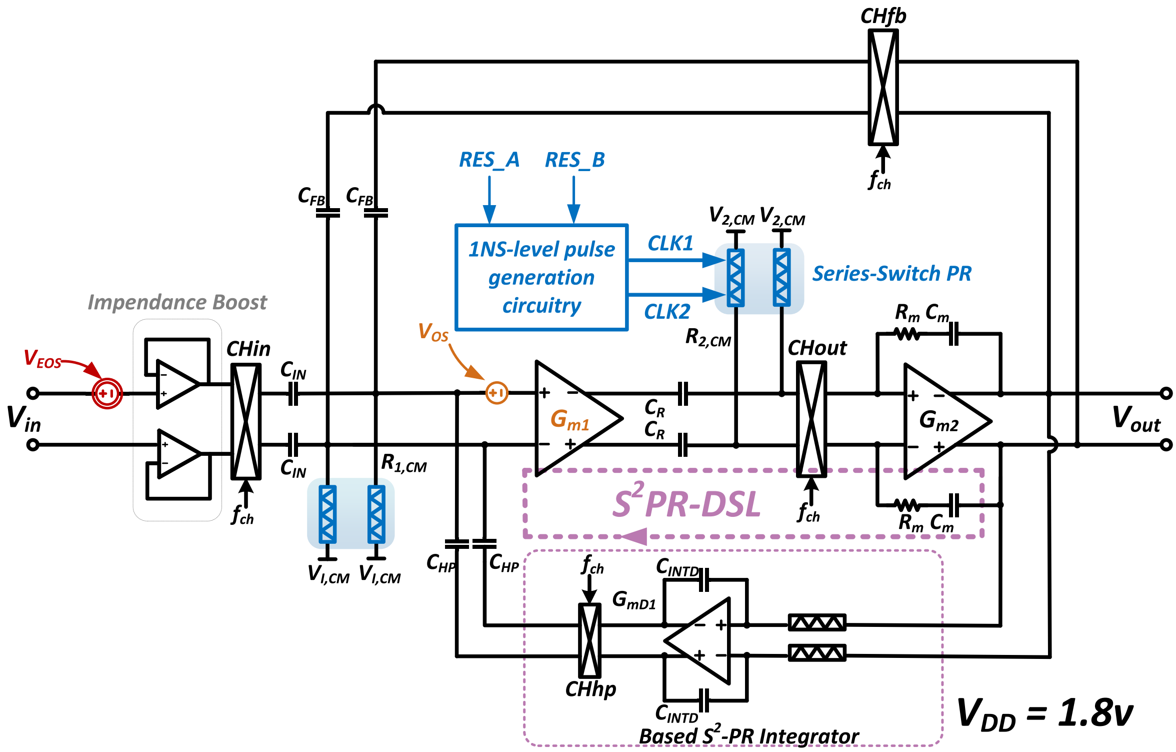

Fig. 2. CCIA system architecture diagram.

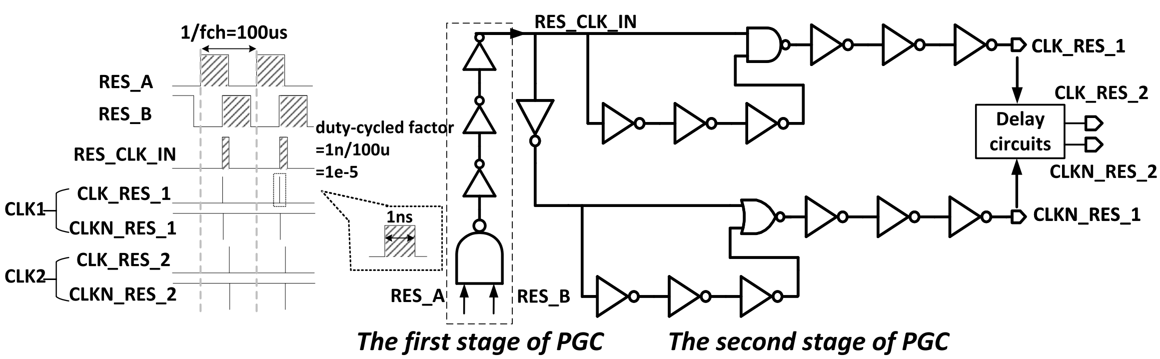

Fig. 3. Pulse generation circuit structure and workflow diagram.

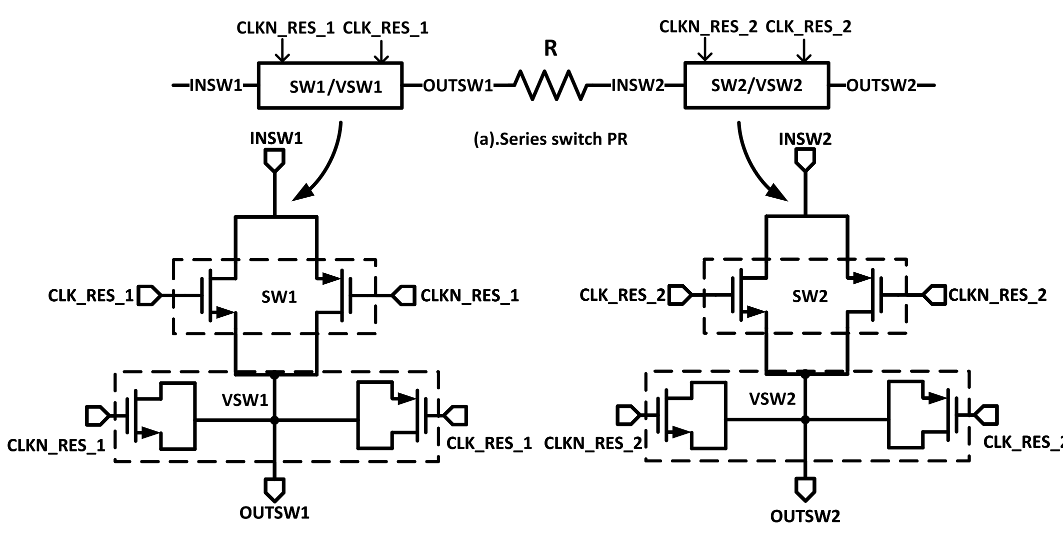

Fig. 4. S^2PR schematic diagram.

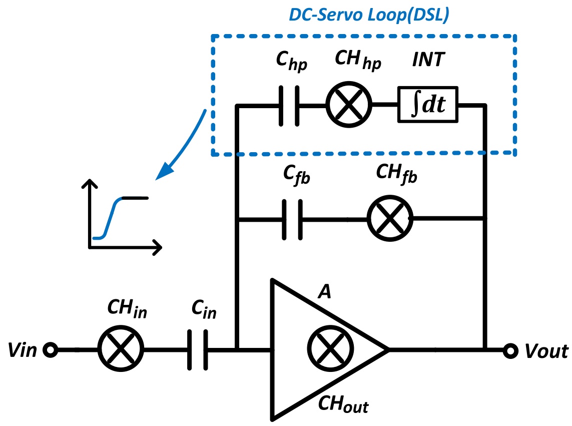

Fig. 5. Block diagram of conventional DSL.

path with an on-time of 1 ns. The amount of charge transferred from the input is very low, so the charge transferred by SW1 will be temporarily stored in the source–drain parasitic capacitance of SW2, which is in its off state now. Then, the CLK1 is flipped again, SW1 is turned off, and the input path is truncated. After a μs-level delay, the CLK2 drives the SW2 to turn on, and the stored charge continues to pass and eventually flows out of the circuit from the output of SW2. So, the minimal charge can flow through this circuit during the 100-μs cycle time so that the equivalent resistance observed at the ends of the circuit during one cycle is very large. In addition, the duty cycles of the CLK1 and CLK2 are controlled by the phase difference between RES_A and RES_B and the frequency F_{res} of the two signals, varying the on-time of S^2PR in one cycle to adjust its equivalent resistance value.

The equivalent resistance value of the S^2PR observed in one cycle is derived as follows, $\Delta\omega$ is the phase difference between signals RES_A and RES_B, and the on-time per cycle of SW1 controlled by CLK1 is t_{on}.

$$t_{on} = \frac{\Delta w\ D}{2\pi\ F_{res}} \tag{2}$$

Both SW1 and SW2 use the CMOS strategy. When the switch is on, the complementary transistors in the switch are in the linear region. The Eq. (3) can be deduced from the relationship between the leakage current in the linear region and the amount of voltage variation in the CMOS switch-resistor model.

$$\Delta vout = \frac{2}{R} - 2\left[\frac{W_p}{L_p}V_{thp} + \frac{W_n}{L_n}(VDD - V_{thn})\right] \tag{3}$$

where Δvout is the voltage difference between the two sides of the resistor R from the beginning to the end of the S^2PR operating state. W_p/L_p and W_n/L_n are the complementary transistor sizes of SW1, respectively, and V_{thp} and V_{thn} are the threshold voltages of the complementary transistors of SW1.

According to the Ohm's law, the equivalent current of the S^2PR in one working cycle can be obtained as:

$$i = \frac{\Delta vout}{R} = \frac{2}{R^2} - \frac{2}{R}\left[\frac{W_p}{L_p}V_{thp} + \frac{W_n}{L_n}(VDD - V_{thn})\right] \tag{4}$$

The equivalent resistance is inversely proportional to the amount of charge observed to flow through the working cycle, and the proposed S^2PR equivalent resistance can be expressed as:

$$Req = \frac{\alpha}{t_{on}\ i} = \frac{2\pi\ F_{res}\ \alpha}{\Delta w\ D} / \left\{\frac{2}{R^2} - \frac{2}{R}\left[\frac{W_p}{L_p}V_{thp} + \frac{W_n}{L_n}(VDD - V_{thn})\right]\right\} \tag{5}$$

where α is the channel charge injection factor, related to the process parameters used in CMOS and inversely proportional to the amount of channel charge injected.

Due to the small amount of charge conducted in one cycle, the channel charge injected by the switch's transistor can greatly affect the planned amount of charge. The CMOS switch structure allows the mutually inverted channel-injected charges to cancel each other out, reducing the channel-injected effect to a large extent. However, since the amount of channel injection charge is related to the transistors' input voltage and threshold voltage, which do not produce equal channel injection charges, the injection cannot be canceled entirely. So, virtual switches VSW1 and VSW2 are added to complement SW1 and SW2, respectively. The structure of the virtual switches is shown in Fig. 4, where the source and drain of VSW1 (VSW2) are connected to the output, and the clock signal driving the virtual switches is inverted with their complementary switches. When SW1 (SW2) is disconnected, VSW1 (VSW2) conducts, and the channel charge deposited on the parasitic capacitance of the switch is absorbed by the virtual switch, which can significantly inhibit channel charge injection into the main path. This not only causes the value of α to be increased, but it also improves the linearity of the S^2PR.

The conventional DSL block diagram shown in Fig. 5 contains the integrator and the chopper CH_{hp} that modulates the DC signal frequency to high [14]. Like CH_{in} and CH_{fb}, these chopper switches are driven by the same clock signal, which ensures that the input signal to each adder in CCIA is in the same band (either all modulated or all at the fundamental frequency). To simplify the calculation, the chopper switches can be identified as wires in the transfer function analysis. Then, the transfer function of the DSL system is:

$$V_{out}(s) * V_{\text{int}}(s) = V_{outi}(s) \tag{6}$$

V_{in}(s) and V_{out}(s) represent the input and output voltages of the CCIA, and V_{int}(s) represent the transfer function of the integrator in the DSL. Moreover, V_{outi}(s) represents the integrator's output voltage. The gain of the amplifier in the integrator is sufficient, and its transfer function is:

$$V_{int}(s) = \frac{1}{s\ C_{dsl}\ R_{dsl}} \tag{7}$$

where C_{dsl} and R_{dsl} are the integrator capacitance and resistance respectively.

The expression for the transfer function in the CCIA established by the voltage relationship between the C_{in}/C_{fb} feedback loop and the DSL loop of C_{hp} is:

$$[V_{in}(s)\ s\ C_{in} - V_{outi}(s)\ s\ C_{hp}]\frac{1}{s\ C_{fb}} = V_{out}(s) \tag{8}$$

Combining Eqs. (6), (7), (8) yields the CCIA system transfer function is:

$$\frac{Vout(s)}{Vin(s)} = \frac{C_{in}\ s}{\frac{C_{hp}}{R_{dsl}\ C_{dsl}} + s\ C_{fb}} \tag{9}$$

It can be observed that a pole is introduced in the CCIA system, which is the high-pass corner introduced by the DC servo loop integrator, and the corner frequency is inversely proportional to C_{dsl} and R_{dsl}.

Considering the effect of the isolation capacitance C_r and the bias resistor $R_{2,CM}$ on the system transfer function, because $R_{2,CM}$ is externally connected to a voltage source given the bias voltage, the final transfer function can be analyzed using the superposition theory:

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{C_{in}\ s}{\left(\frac{C_{hp}}{R_{dsl}\ C_{dsl}} + s\ C_{fb}\right)\left(1 + \frac{1}{R_{2,CM}C_r\ s}\right)} \tag{10}$$

Since the bias resistor $R_{2,CM}$ has the S^2PR structure, which has a very large equivalent resistance, the pole introduced by C_r and $R_{2,CM}$ is very close to the zero point. It has a negligible effect on analyzing the transmission characteristics in the desired frequency band.

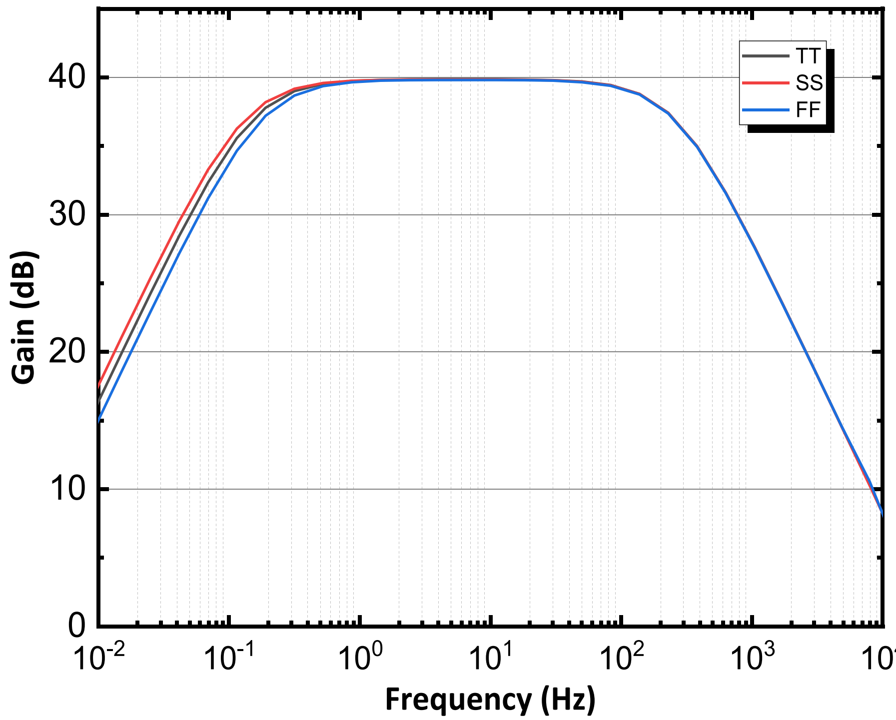

Fig. 6. System amplitude–frequency characteristics at different process corners.

The size of C_{hp} is determined by the amplitude of the electrode DC offset voltage.

$$C_{hp} = \frac{C_{in} V_{eo}}{V_{outmax}} \tag{11}$$

where V_{eo} is the highest DC offset voltage that can be suppressed. Set C_{hp} to 1 pF and V_{outmax} to 1.8 V power supply voltage, and calculate V_{eo} that can reach up to 180 mV.

After determining the C_{hp}, Eq. (12) can be calculated based on Eq. (8):

$$f_{hp} = \frac{C_{hp}}{C_{fb}} f_{0DSL} \tag{12}$$

The high-pass corner frequency of the integrator in DSL is denoted by f_{0DSL}. f_{hp} is set to 0.2 Hz initially, which is brought into Eq. (12) to obtain the high-pass corner frequency of the integrator calculated to be 0.05 Hz. In summary, an integrator with such a low high-pass corner frequency severely demands the design effort. The passive device will occupy a considerable area if a conventional RC integrator is used. In this work, the S^2PR technique is applied to realize very large time constant integrators.

3. Simulation results

The proposed CCIA is realized by a standard 0.18-μm CMOS process. The amplitude–frequency characteristics of CCIA at different process corners are shown in Fig. 6. The designed S^2PR-DSL strongly rejects differential DC voltage inputs. The in-band gain was set to 40 dB initially, and due to parasitic capacitive-resistive effects, the band gain observed in the actual simulation is 39.8 dB. It can be observed that the S^2PR-DSL introduces a 0.159-Hz high-pass corner in the system at the TT corner, 0.138 Hz at the SS corner, and 0.18 Hz at the FF corner. It indicates that the S^2PR-DSL achieves the desired performance and that the high-pass corner frequency maintains a stable action at all process corners, showing the S^2PR structure with high linearity. Moreover, it consumes a tiny area to produce a giga-omega equivalent resistance, which lays the technical foundation for the fully integrated realization of DSL.

The CCIA uses chopper technology to reduce the input-referred noise effectively. The spectral density of the input-referred noise of the CCIA in the 0.5 Hz–150 Hz ECG signal band is shown in Fig. 7, with the in-band integration noise of 1.78 μV_{rms} at the TT corner, 1.83 μV_{rms} at the FF corner, and 1.76 μV_{rms} at the SS corner. The integrated noise at all process corners is less than typical for the ECG signal, so the ECG signal will not be drowned out by the noise of the CCIA, which ensures that the designed CCIA can accomplish the high-precision acquisition of the ECG signal.

The input impedance after adding the buffer is shown in Fig. 8. The input impedance of the input signal at 50 Hz is 772.2 MΩ, which is about 340 times higher than the input impedance of the chopper switch as input. As the input signal frequency increases to kHz, the input impedance of the impedance boost circuit tends to decrease; this is because of the parasitic capacitance of the buffer input transistor gate to source–drain terminals. As the signal frequency increases, the parasitic capacitance equivalent input impedance decreases, leading to another change in the signal transfer path and a decrease in the input equivalent impedance. However, for an ECG signal with a frequency band of 0.5 Hz–150 Hz, the impedance observed at the input signal frequency of 150 Hz is 392 MΩ, which is also enough for ECG signal acquisition.

A sinusoidal signal with a frequency of 9.033 Hz and a V_{pp} of 10 mV is applied to the CCIA input. Fast Fourier Transform analyzes the output signal to obtain the spectrum Fig. 9. It can be observed that the fundamental frequency corresponds to 9.033 Hz, whose amplitude is 0.502 V. It is shown that its second, third, and fifth harmonics are relatively large in amplitude, respectively 0.1061 mV, 0.3648 mV, 0.1611 mV. The total harmonic distortion (THD) is calculated to be −61.6 dB. Such a low THD proves that the designed CCIA can realize the high-quality transmission of weak signals.

The two-stage PGC generates ns-level pulse width signals CLK1 and CLK2 to drive the S^2PR with a period of 100 μs. The observed pulse widths are 1.1004 ns at the TT corner, 1.013 ns at the FF corner, and 1.221 ns at the SS corner. The actual duty cycle matches the design value of about 1/100 000, and the equivalent resistance value of the S^2PR achieves more than 27.5 GΩ at all process corners and temperatures, which meets the design expectation.

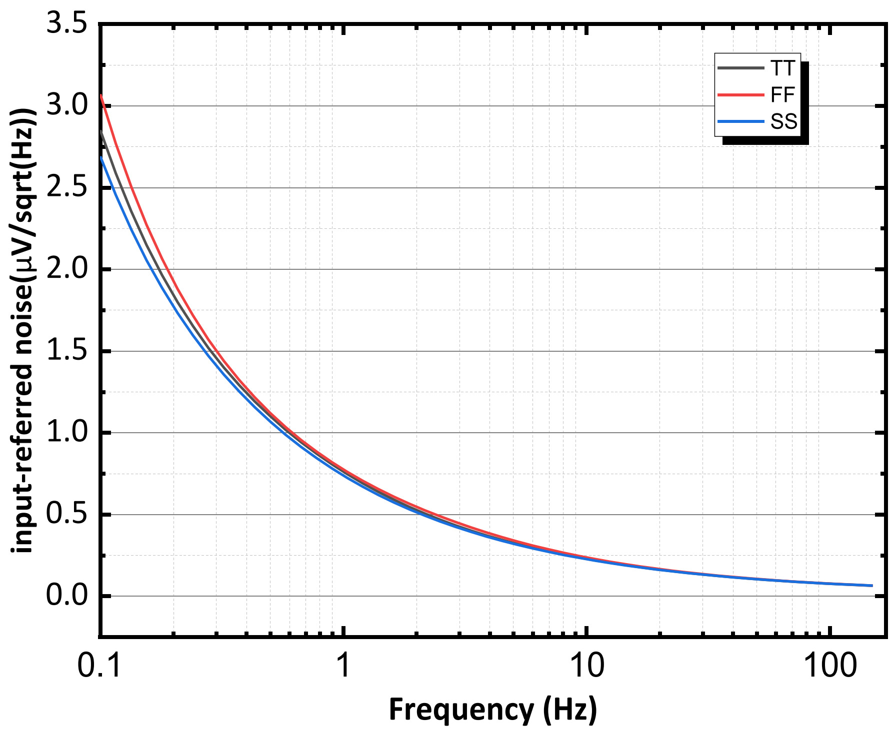

Fig. 7. Input-referred noise spectrum density within the ECG-signal bandwidth.

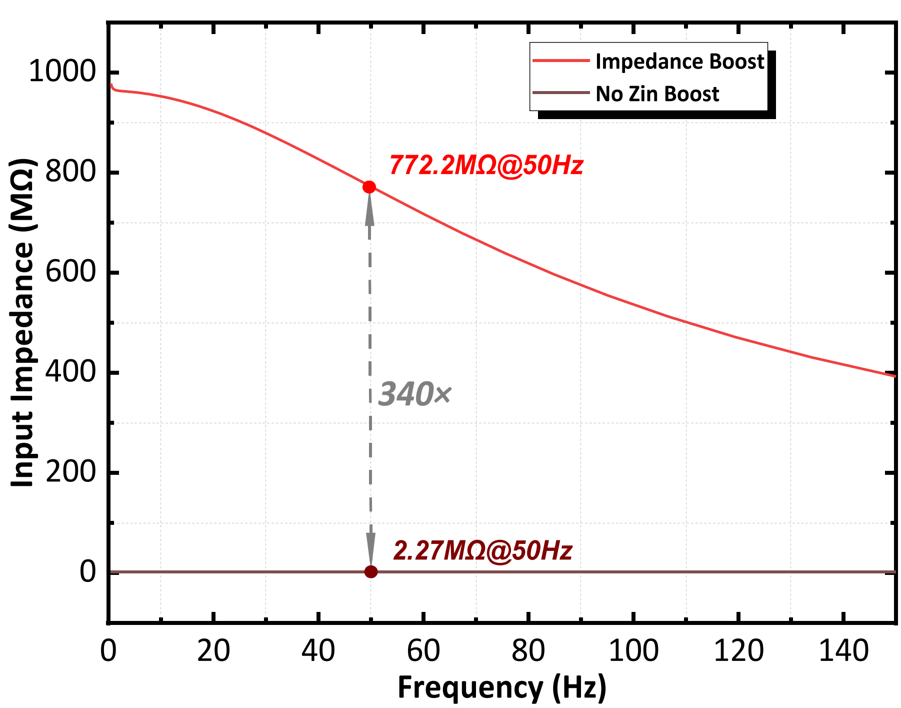

Fig. 8. Input resistance in ECG signal band before and after impedance boost.

The total power consumption of the core circuit including auxiliary circuits (e.g., bias circuit) is 0.54 mW, which is normal in high precision applications. The power consumption of the S^2PR-DSL including the two-stage PGC is only 92.7 μW, and the overall performance of the proposed S^2PR-DSL is highly competitive.

4. Conclusion

In this work, an S^2PR-DSL method is proposed to eliminate the DC offset voltage introduced by the sampling electrodes during ECG acquisition. The circuit can avoid the instrumentation amplifier output voltage saturation with a small active area. A high-pass corner of

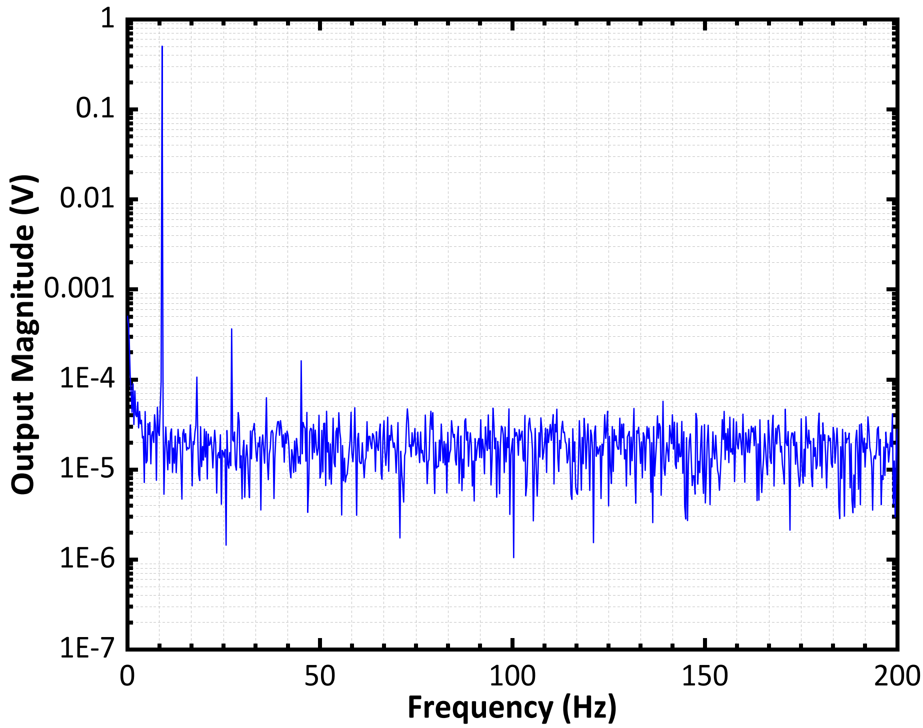

Fig. 9. CCIA linearity simulation result.

Table 1
Performance summary and comparison with prior works.

Parameters	[15][a]	[16]	[17][a]	[18]	[19][b]	[20]	**This work**[b]
Architecture	CCIA	CFIA	CFIA	CCIA	CCIA	CFIA	CCIA
Technology	65 nm	0.18 μm	0.18 μm	0.18 μm	0.18 μm	0.18 μm	**0.18 μm**
Gain	0–24.86 dB	40 dB	40–60 dB	37 dB	34–52 dB	45–63 dB	**39.8 dB**
Electrode DC offset tolerance	N.I.	250 mV	N.I.	N.I.	N.I.	N.I.	**180 mV**
Supply voltage	0.6 V	1.8 V	1.8 V	1.5 V	1.8 V	1 V	**1.8 V**
Input-referred noise	6.628 μV_{rms} (1–100 Hz)	3.8 μV_{rms} 0.45–100 Hz)	9.56 μV_{rms} (0.05–250 Hz)	11.8 μV_{rms} (1–300 Hz)	1.36 μV_{rms} (0.5–100 Hz)	3L.16 μV_{rms} (1–10k Hz)	**1.78 μV_{rms} (0.5–150 Hz)**
Input impendence	N.I.	N.I.	N.I.	N.I.	550 MΩ@100 Hz	N.I.	**772 MΩ@50 Hz**
High-pass corner frequency	3 Hz	0.45 Hz	N.I.	0.8 Hz	<0.5 Hz	0.5 Hz	**0.159 Hz**

N.I. stands for not included.
[a] Post-simulation Results.
[b] Pre-simulation Results.

0.16 Hz is introduced into the instrumentation amplifier system at the TT corner. In addition, the proposed instrumentation amplifier achieves high input impedance and excellent linearity. The DC offset caused by non-ideal factors is suppressed without affecting the high-precision amplification of the weak signals in the desired frequency band. The performance of the proposed instrumentation amplifier is summarized in Table 1 and is compared with the state-of-art. The proposed instrumentation amplifier is ideally suited for systems that acquire ECG signals and other related applications.

CRediT authorship contribution statement

Jinze Song: Methodology, Investigation, Writing – original draft, Writing – review & editing. **Kui Wen:** Conceptualization, Writing – review & editing. **Ruixue Ding:** Data curation. **Min Chen:** Supervision. **Shubin Liu:** Formal analysis. **Zhangming Zhu:** Visualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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References

[1] Tingyu Wang, Kui Wen, Longjie Zhong, Shubin Liu, A 0.26 μV_{rms} instrumentation amplifier based on discrete-time DC servo loop with successive-approximation-compensation technique and Chopper DAC Array, Microelectron. J. 139 (2023) 105906.

[2] R.H. Olsson, D.L. Buhl, A.M. Sirota, G. Buzsaki, K.D. Wise, Band-tunable and multiplexedintegrated circuits for simultaneous recording and stimulation with microelectrodearrays, IEEE Trans. Biomed. Eng. 52 (2005) 1303–1310.

[3] W. Wattanapanitch, M. Fee, R. Sarpeshkar, An energy-efficient micropower neural recordingamplifier, IEEE Trans. Biomed. Circuits Syst. 1 (2007) 136–147.

[4] M. Chae, W. Liu, Z. Yang, T. Chen, J. Kim, M. Sivaprakasam, et al., A 128-Channel 6 MW Wireless Neural RecordingIC with on-the-Fly Spike Sorting and UWB Transmitter, IEEE ISSCC Dig. Tech. Papers, 2008, pp. 146–603.

[5] X. Zou, X. Xu, L. Yao, Y. Lian, A 1-v 450-nW fully integrated programmable biomedical sensor interface chip, IEEE J. Solid-State Circuits 44 (4) (2009) 1067–1077.

[6] Y. Perelman, R. Ginosar, An integrated system for multichannel neuronal recording with spike/LFP separation, integrated A/D conversion and threshold detection, IEEE Trans. Biomed. Eng. 54 (1) (2007) 130–137.

[7] T. Horiuchi, T. Swindell, D. Sander, P. Abshire, A low-power CMOS neural amplifier with amplitude measurements for spike sorting, in: 2004 IEEE International Symposium on Circuits and Systems, ISCAS, 2004, pp. IV–29.

[8] Q. Fan, F. Sebastiano, J.H. Huijsing, K.A.A. Makinwa, A 1.8 μW 60 nV/$\sqrt{\text{Hz}}$ capacitively-coupled chopper instrumentation amplifier in 65 nm CMOS for wireless sensor nodes, IEEE J. Solid-State Circuits 46 (7) (2011) 1534–1543.

[9] Chengying Chen, Liming Chen, Xinghua Wang, Feng Zhang, A 0.6V, 8.4μW AFE circuit for biomedical signal recording, Microelectron. J. 75 (2018) 105–112.

[10] Y. Huang, J. Liu, L. Cheng, A, 1.7μW/channel time-division-multiplexing analog front-end for bio-potential recording, IEEE Trans. Circuits Syst. II http://dx.doi.org/10.1109/TCSII.(2024).3367636.

[11] H. Chandrakumar, D. Marković, A high dynamic-range neural recording chopper amplifier for simultaneous neural recording and stimulation, IEEE J. Solid-State Circuits 52 (3) (2017) 645–656.

[12] K. Wen, S. Liu, L. Zhong, Y. Shen, Z. Zhu, A −64.3 dB THD, 26 nV/$\sqrt{\text{Hz}}$ bio-potential readout analog-front-end amplifier with a Gm-C integrator-implanted DC servo loop, and a bulk-driven ripple reduction loop, IEEE Trans. Circuits Syst. I. Regul. Pap. 71 (2) (2024) 537–547.

[13] L. Yan, J. Bae, Challenges of physiological signal measurements using electrodes: Fudamentals to understand the instrumentation, IEEE Solid-State Circuits Mag. 9 (4) (2017) 90–97.

[14] U. Ha, J. Lee, M. Kim, T. Roh, S. Choi, H.-J. Yoo, An EEG-NIRS multimodal SoC for accurate anesthesia depth monitoring, IEEE J. Solid-State Circuits 53 (6) (2018) 1830–1843.

[15] Siwan Dong, Ruoyu Zhang, Chuqiang Jing, Menghan Yuan, A 0.6V 119 dB high-CMRR low-NEF PGA with common-mode voltage control for ECG recording, Microelectron. J. 146 (2024) 106159.

[16] Z. Hoseini, M. Nazari, K.-S. Lee, H. Chung, Current feedback instrumentation amplifier with built-in differential electrode offset cancellation loop for ECG/EEG sensing frontend, IEEE Trans. Instrum. Meas. 70 (2020) 1–11.

[17] P.K. Jha, P. Patra, A. Dutta, g_m/I_D sizing and analysis of a recycling folded-cascode OTA for ECG signal conditioning in $0.18\mu\text{m}$ CMOS technology, Analog Integr. Circ. Sig. Process. 115 (2023) 263–278.

[18] Shashank Dwivedi, Anup K. Gogoi, A compact sub-Hertz local field potential amplifier for implantable biomedical devices, Microelectron. J. 128 (2022) 105539.

[19] K. Wen, Y. Shen, T. Wang, et al., A low-noise high-precision analog front end system for biopotential signals recording application, Analog Integr. Circ. Sig. Process. 111 (2022) 103–115.

[20] Luat Tran, Hyouk-Kyu Cha, An ultra-low-power neural signal acquisition analog front-end IC, Microelectron. J. 107 (2021) 104950.